

SOFTWARE REQUIREMENT SPECIFICATION

Project Name (Code)

– Hanoi, Oct 2025 –

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I. Record of Changes

Date	A* M, D	In charge	Change Description

*A - Added M - Modified D - Deleted

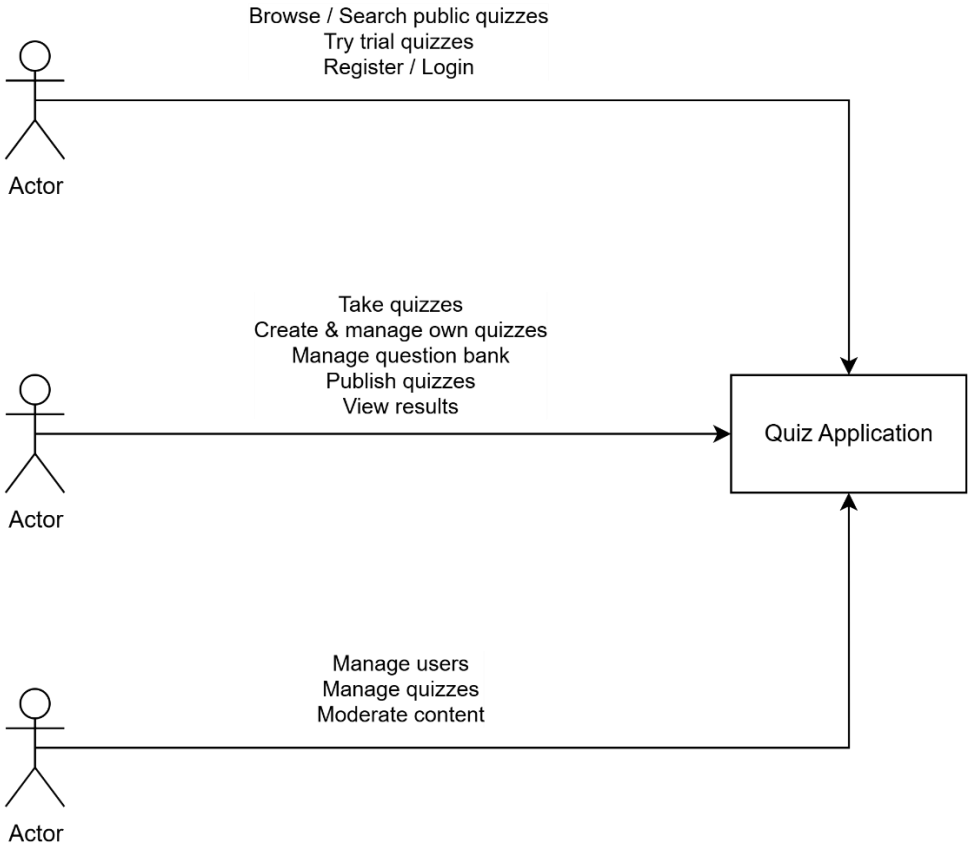
II. Software Requirement Specification

1. Overall Requirements

1.1 Context Diagram

Quiz Application is a system that allows users to discover, create, manage, and participate in quizzes. Users can create their own quizzes and questions, maintain a personal question bank, reuse public questions, and publish quizzes for other users to access.

The system supports three main actors: **Guest, User, and Administrator**. Guests can browse public quizzes and try a limited number of trial quizzes. Registered users can participate in quizzes, create and manage their own quizzes and question banks. Administrators are responsible for managing users, quizzes, questions, and inappropriate content within the system.

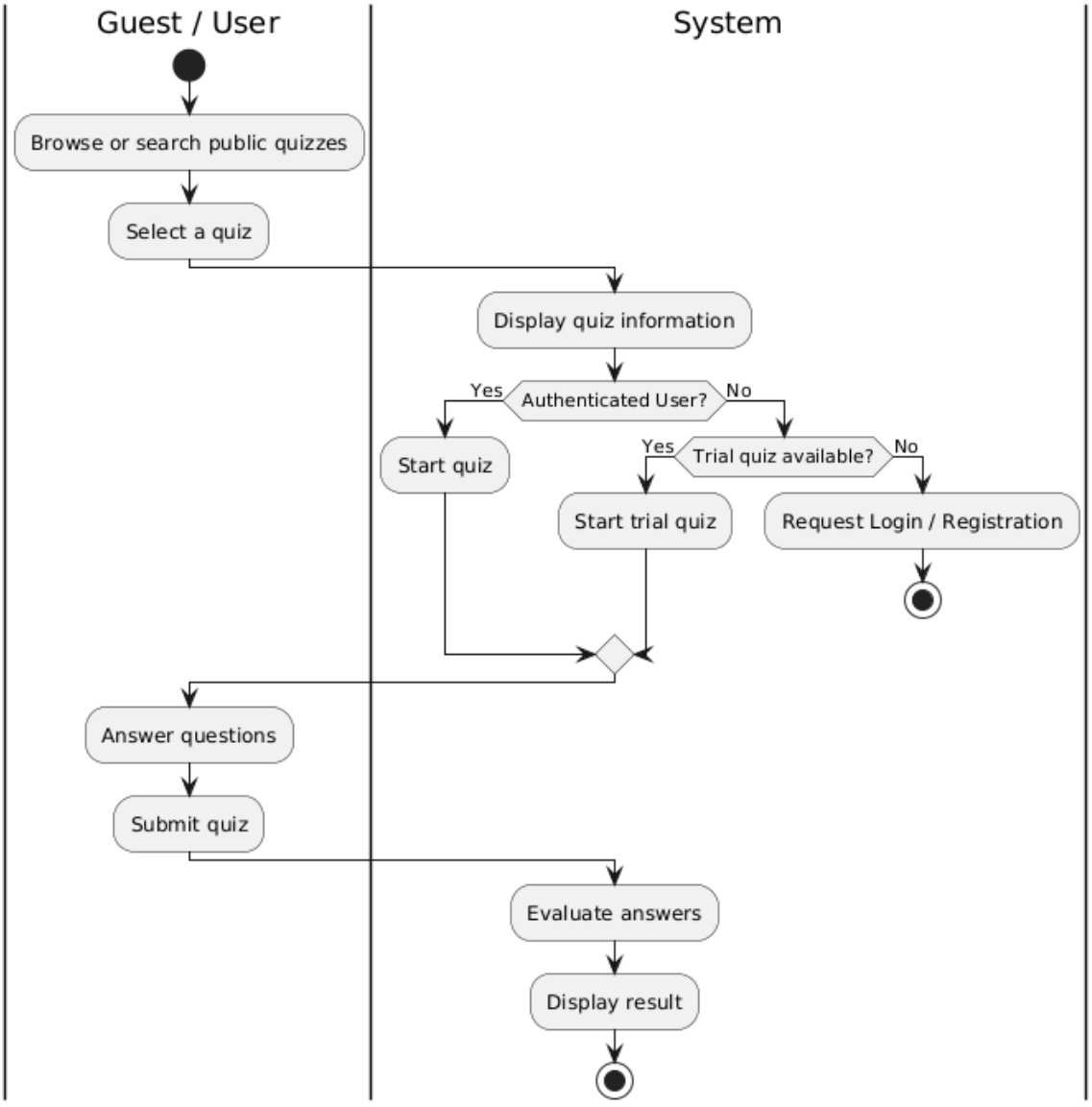


1.2 Main Business Processes

The Quiz Application supports the main business processes of participating in quizzes, creating and publishing quizzes, and administering system content. Guests have limited access to trial quizzes, while registered users can fully participate in quizzes and create their own quiz content.

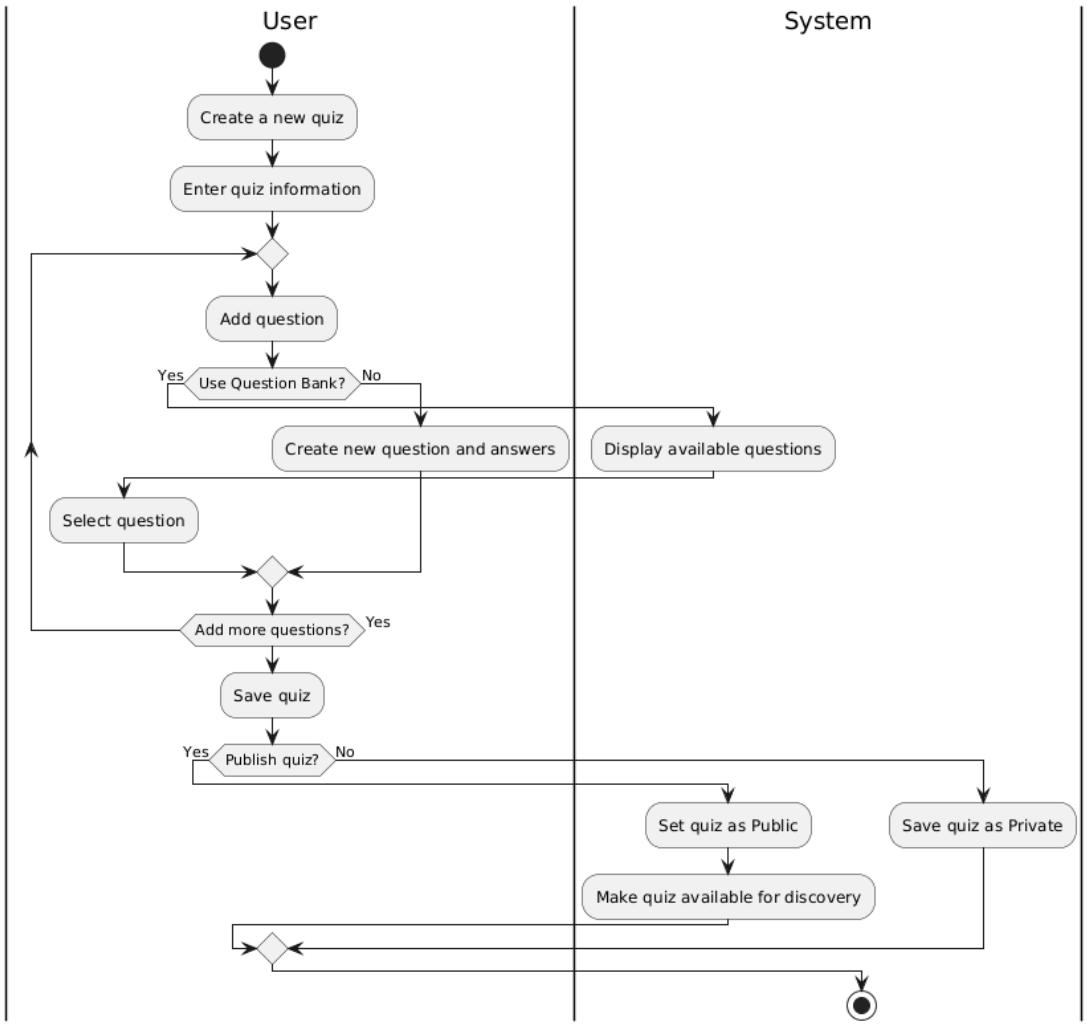
1.2.1 Quiz Participation Process

This process describes how a **Guest or User searches for and participates in a public quiz**. Guests can only access a limited number of trial quizzes. When the trial limit is reached, the Guest must register or log in to continue. Registered Users can participate in available public quizzes and view their results after submission.



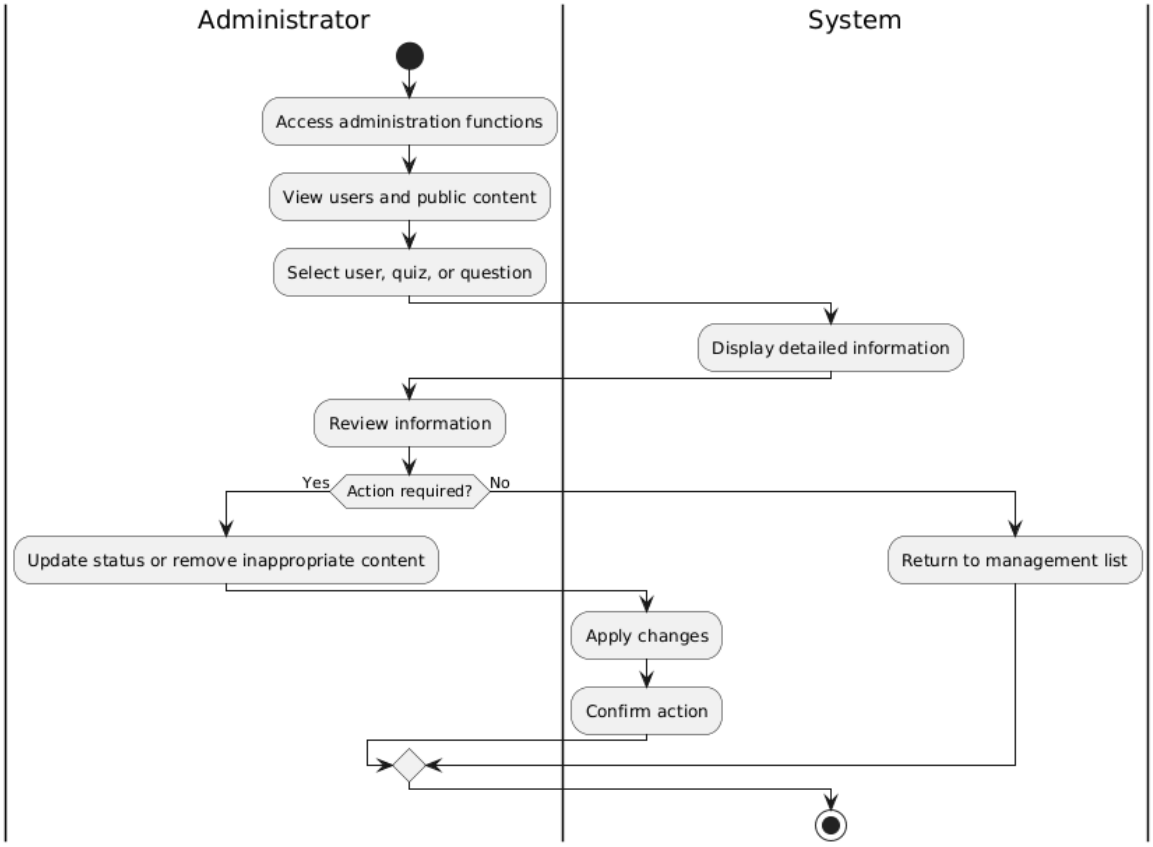
1.2.2 Quiz Creation and Publishing Process

This process allows a registered **User to create and manage a quiz**. Questions can be created manually or selected from the Question Bank. After completing the quiz, the User can save it as private or publish it for other users.



1.2.3. Content Administration Process

This process describes how an **Administrator monitors and manages users and public content**. The Administrator can review quizzes or questions and take appropriate action when inappropriate content is identified.



1.3 User Requirements

1.3.1 Actors

The Quiz Application has three main actors: **Guest, User, and Administrator**. Each actor has different permissions based on their authentication status and role in the system.

#	Actor	Description
1	Guest	An unauthenticated visitor who can browse and search public quizzes, view quiz information, and try a limited number of trial quizzes. A Guest must register or log in to access additional features.
2	User	A registered and authenticated user who can participate in public quizzes, view results, create and manage their own quizzes, manage questions in their Question Bank, and publish quizzes for other users.
3	Administrator	An authenticated user with administrative privileges who manages users and system content. The Administrator can review and manage quizzes and questions, moderate inappropriate content, and maintain the overall system.

1.3.2 Use Cases (UC)

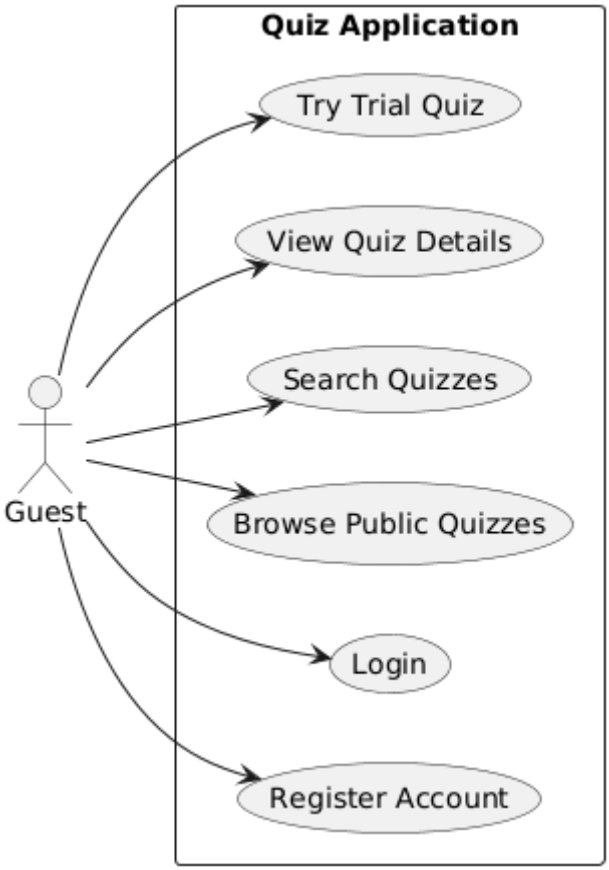
The following use cases describe the main interactions between the actors and the Quiz Application.

ID	Use Case	Feature	Use Case Description
UC01	Register Account	Authentication	Guest creates a new user account to access the full features of the system.
UC02	Login	Authentication	Guest or registered user authenticates with the system using valid credentials.
UC03	Logout	Authentication	Authenticated User or Administrator logs out of the system.
UC04	Browse Public Quizzes	Quiz Discovery & Participation	Guest or User views quizzes that have been published for public access.
UC05	Search Quizzes	Quiz Discovery & Participation	Guest or User searches for public quizzes using keywords or available search criteria.
UC06	View Quiz Details	Quiz Discovery & Participation	Guest or User views basic information about a selected public quiz before attempting it.
UC07	Try Trial Quiz	Quiz Discovery & Participation	Guest attempts an available trial quiz subject to the Guest usage limitation.
UC08	Take Quiz	Quiz Discovery & Participation	Authenticated User participates in an available public quiz and answers its questions.
UC09	Submit Quiz	Quiz Discovery & Participation	User submits completed answers for evaluation.
UC10	View Quiz Result	Quiz Discovery & Participation	User views the result of a submitted quiz.
UC11	Create Quiz	Quiz Management	User creates a new quiz and provides its basic information.
UC12	Manage Own Quiz	Quiz Management	User views, updates, or deletes quizzes that they created.
UC13	Manage Quiz Questions	Quiz Management	User adds, updates, removes, or arranges questions and answers within their own quiz.
UC14	Publish Quiz	Quiz Management	User publishes their quiz for other users to discover and participate in, or changes it back to private.
UC15	Manage Question Bank	Question Bank	User creates, updates, deletes, and manages questions stored in their personal Question Bank.
UC16	Search Public Questions	Question Bank	User searches for reusable questions that are available publicly in the Question Bank.
UC17	Reuse Question	Question Bank	User selects an existing question from the Question Bank and adds it to their quiz.
UC18	Manage Users	Administration	Administrator views and manages registered user accounts and their statuses.
UC19	Manage System Quizzes	Administration	Administrator reviews and manages quizzes available in the system.
UC20	Moderate Content	Administration	Administrator reviews and takes action against inappropriate quizzes or questions.

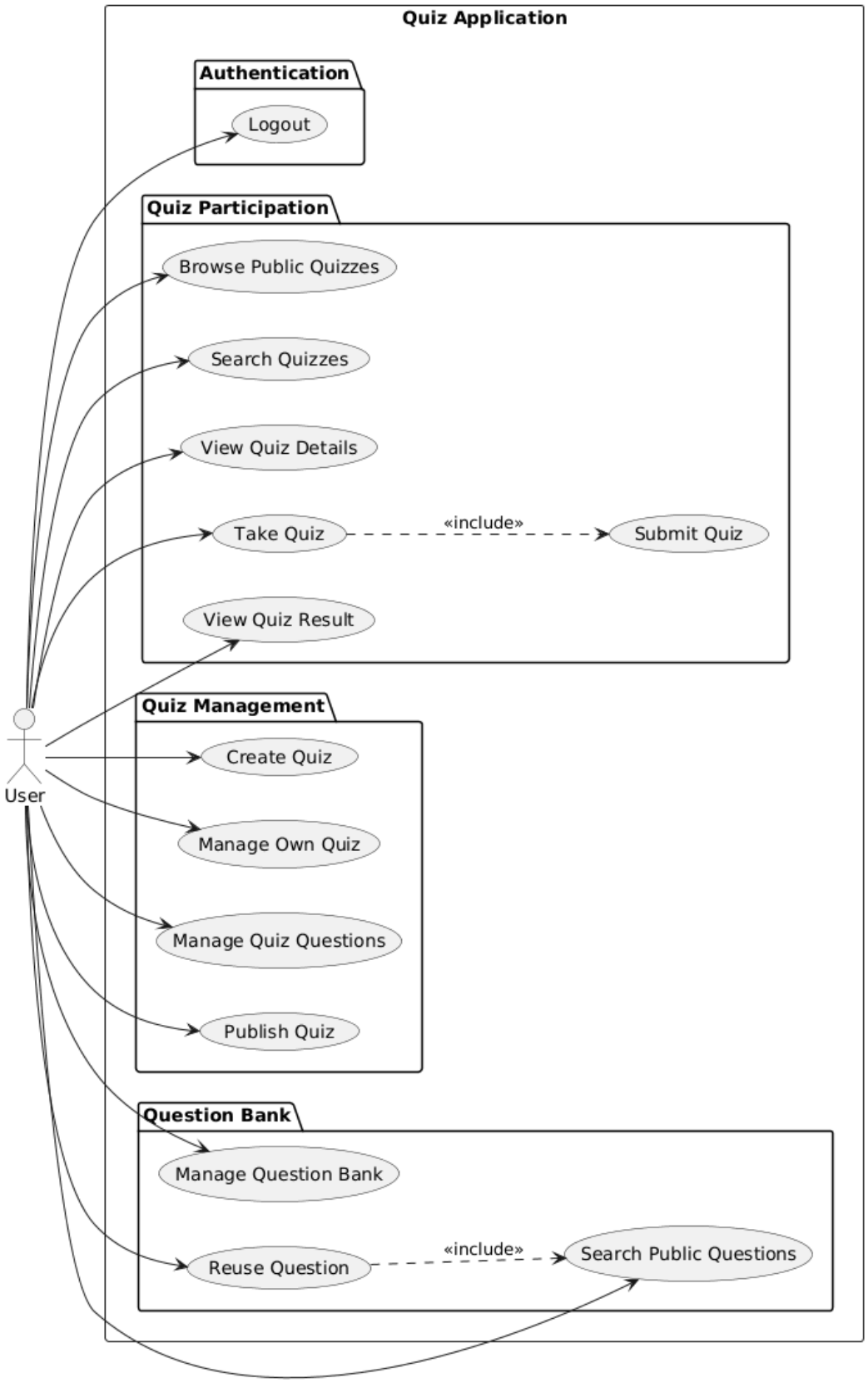
1.3.3 Use Case Diagrams

The following diagrams illustrate the interactions between the main actors and the Quiz Application. The use cases are grouped by actor to provide a clear view of the functionalities available to each user role.

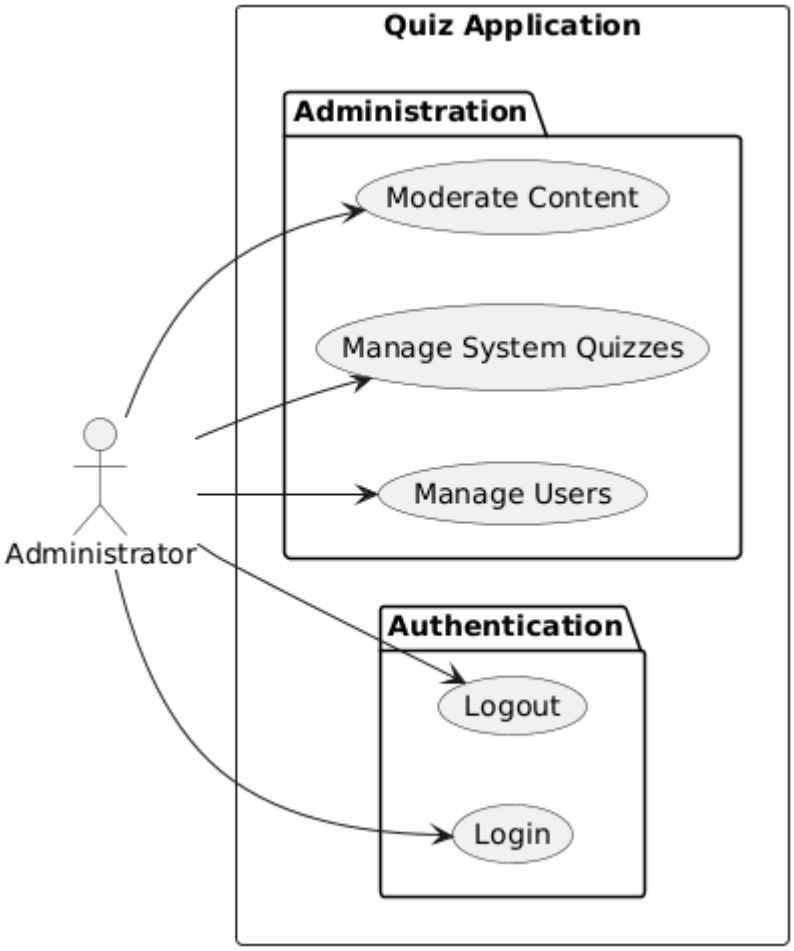
1.3.2.1 UCs for Guest



1.3.2.2 UCs for User



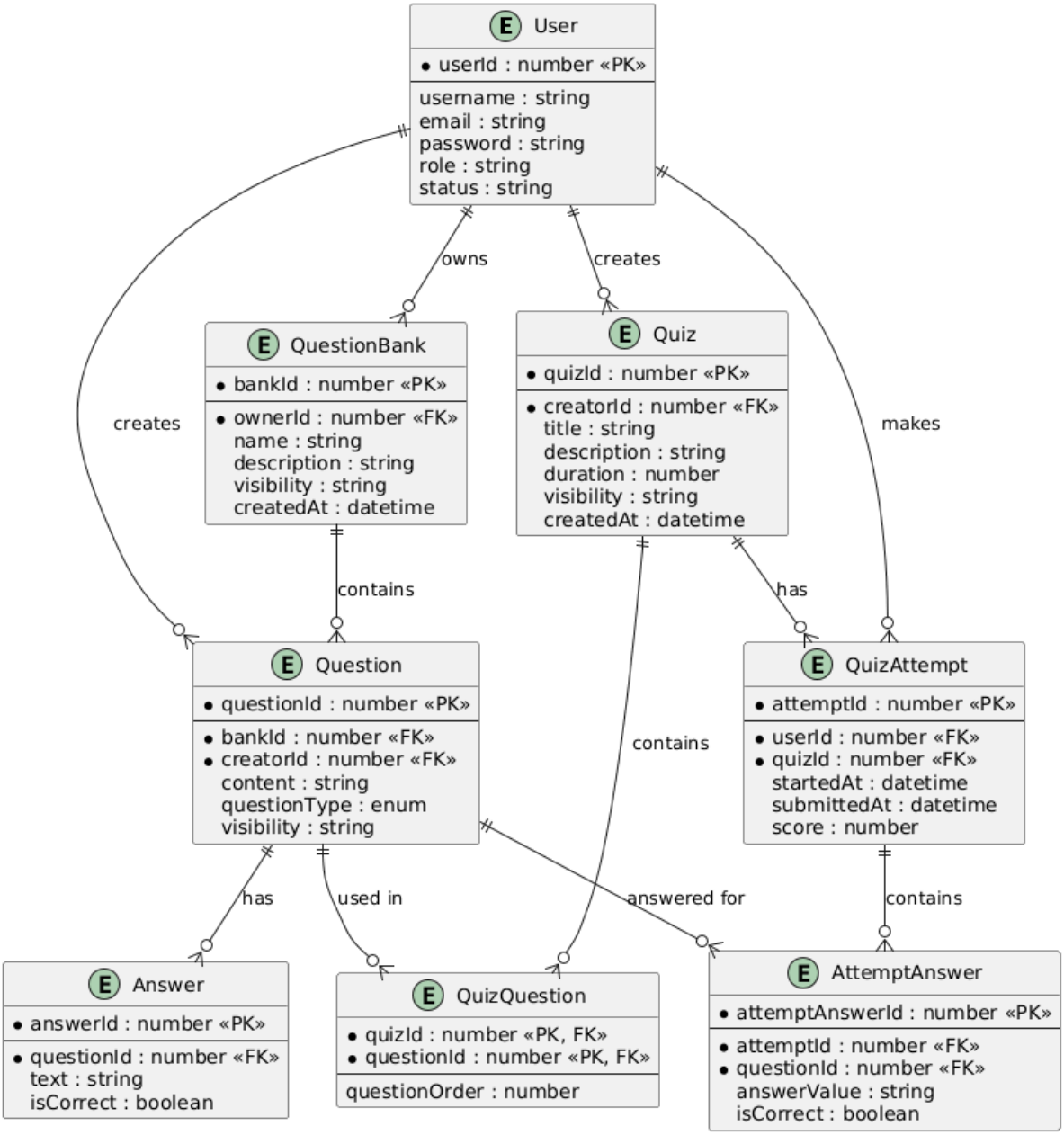
1.3.2.3 UCs for Administrator



1.4 Entity Relationship Diagram

The Entity Relationship Diagram describes the main data entities and their relationships within the Quiz Application. In addition to quizzes, questions, and answers, the system provides Question Banks that allow users to organize, store, and reuse questions when creating quizzes.

Questions stored in a Question Bank can be reused across multiple quizzes. The system also maintains quiz attempts and submitted answers to support quiz participation and result evaluation.



Entities Description

#	Entity	Description
1	User	Stores registered user information and roles. Users can create quizzes, own Question Banks, and participate in quizzes.
2	QuestionBank	Stores and organizes reusable questions owned by a User. A Question Bank may be private or publicly accessible for question reuse.
3	Quiz	Stores quiz information such as title, description, duration, visibility, and creator.
4	Question	Stores question content and type. A question belongs to a Question Bank and can be reused in multiple quizzes.
5	Answer	Stores possible or expected answers associated with a Question and identifies correct answers when applicable.

6	QuizQuestion	Associates Questions with Quizzes and maintains their order, enabling the same Question to be reused across multiple Quizzes.
7	QuizAttempt	Stores information about a User's attempt at a Quiz, including start time, submission time, and score.
8	AttemptAnswer	Stores answers submitted for individual Questions during a Quiz attempt.

2. Use Case Specifications

The following section provides detailed specifications for the main use cases of the Quiz Application. The use cases are grouped by major system features. Simple CRUD and data-viewing use cases are described in the Functional Requirements section.

2.1 Authentication

2.1.1 Register Account

Item	Description
Use Case ID	UC01
Use Case Name	Register Account
Primary Actor	Guest
Secondary Actor	None
Description	As a Guest, I want to create an account so that I can access the full features of the Quiz Application.
Preconditions	The Guest is not authenticated and does not have an active account using the same email/username.
Postconditions	A new User account is created successfully and can be used to log in to the system.
Normal Sequence / Flow	1. Guest selects **Register**. 2. System displays the registration form. 3. Guest enters required account information. 4. Guest submits the registration form. 5. System validates the entered information. 6. System creates a new User account. 7. System displays a successful registration message and redirects the Guest to Login.
Alternative Sequences / Flows	**A1 – Required information is missing:** System displays validation messages and asks the Guest to complete the required fields. **A2 – Email or username already exists:** System rejects the registration and informs the Guest to use another email/username. **A3 – Invalid input:** System displays an appropriate validation message and does not create the account.

2.1.2 Login

Item	Description
Use Case ID	UC02
Use Case Name	Login
Primary Actor	Guest / User / Administrator
Secondary Actor	None
Description	As a registered user, I want to log in to the system so that I can access functions according to my role and permissions.
Preconditions	The user has a valid account in the system and is currently not authenticated.
Postconditions	The user is authenticated successfully and redirected to the appropriate system area based on the account role.
Normal Sequence / Flow	1. User selects **Login**. 2. System displays the login form. 3. User enters email/username and password. 4. User submits the login request. 5. System validates the credentials. 6. System identifies the user's role. 7. System creates an authenticated session. 8. System redirects the user to the appropriate page.
Alternative Sequences / Flows	**A1 – Required fields are empty:** System displays validation messages. **A2 – Incorrect credentials:** System denies access and displays an authentication error. **A3 – Account is inactive or blocked:** System denies login and informs the user of the account status.

2.2 Quiz Discovery and Participation

2.2.1 Try Trial Quiz

Item	Description
Use Case ID	UC07
Use Case Name	Try Trial Quiz
Primary Actor	Guest
Secondary Actor	None
Description	As a Guest, I want to try a limited public quiz so that I can experience the Quiz Application before creating an account.

Item	Description
Preconditions	The Guest is not authenticated. The selected quiz is available for trial access.
Postconditions	The Guest completes or exits the trial quiz. The system records the trial usage for limitation control if applicable.
Normal Sequence / Flow	1. Guest browses or searches available public quizzes. 2. Guest selects a trial-enabled quiz. 3. System checks whether the Guest is still allowed to take a trial quiz. 4. System displays the quiz questions. 5. Guest answers the questions. 6. Guest submits the quiz. 7. System evaluates the submitted answers. 8. System displays the trial result.
Alternative Sequences / Flows	**A1 – Trial limit reached:** System does not allow the Guest to continue and requests registration or login. **A2 – Quiz is no longer available:** System informs the Guest that the quiz cannot be accessed. **A3 – Guest exits before submission:** The attempt is not evaluated and the Guest returns to the quiz list or details page.

2.2.2 Take Quiz

Item	Description
Use Case ID	UC08
Use Case Name	Take Quiz
Primary Actor	User
Secondary Actor	None
Description	As a User, I want to participate in a public quiz so that I can answer questions and evaluate my knowledge.
Preconditions	The User is authenticated. The selected Quiz is available for participation.
Postconditions	A Quiz Attempt is created and the User's answers can be submitted for evaluation.
Normal Sequence / Flow	1. User browses or searches public quizzes. 2. User selects a quiz. 3. System displays quiz information. 4. User selects **Start Quiz**. 5. System creates a new Quiz Attempt and displays the quiz questions. 6. User answers the questions. 7. User reviews the entered answers if required.

Item	Description
	8. User selects **Submit Quiz**. 9. System continues to the Submit Quiz use case.
Alternative Sequences / Flows	**A1 – Quiz is unavailable:** System informs the User that the quiz cannot be started. **A2 – User exits before submission:** System ends or temporarily leaves the current attempt according to the system policy. **A3 – Quiz duration expires:** System automatically ends the attempt and submits available answers if time limitation is enabled.

2.2.3 Submit Quiz

Item	Description
Use Case ID	UC09
Use Case Name	Submit Quiz
Primary Actor	User
Secondary Actor	None
Description	As a User, I want to submit my quiz answers so that the system can evaluate my attempt and provide a result.
Preconditions	The User has an active Quiz Attempt and has started the selected Quiz.
Postconditions	The Quiz Attempt is marked as submitted. The submitted answers and calculated result are stored by the system.
Normal Sequence / Flow	1. User selects **Submit Quiz**. 2. System receives the answers from the current Quiz Attempt. 3. System validates the submitted data. 4. System compares the submitted answers with the correct answers where automatic evaluation is supported. 5. System calculates the quiz result. 6. System stores the submitted answers and result. 7. System marks the Quiz Attempt as completed. 8. System displays the result to the User.
Alternative Sequences / Flows	**A1 – Some questions are unanswered:** System may warn the User before final submission and allow the User to return to the quiz. **A2 – Submission data is invalid:** System rejects the submission and displays an appropriate error message. **A3 – Question type requires manual evaluation:** System stores the submitted answer and marks the relevant question as pending evaluation rather than assigning an immediate final score.

2.3 Quiz Management

2.3.1 Create Quiz

Item	Description
Use Case ID	UC11
Use Case Name	Create Quiz
Primary Actor	User
Secondary Actor	None
Description	As a User, I want to create a quiz so that I can organize questions and make the quiz available for my own use or for other users.
Preconditions	The User is authenticated and has permission to create quizzes.
Postconditions	A new Quiz is created and associated with the User as its creator. The Quiz is initially saved as private or draft until the User publishes it.
Normal Sequence / Flow	1. User selects **Create Quiz**. 2. System displays the quiz creation form. 3. User enters quiz information such as title, description, and duration. 4. User adds questions manually or selects existing questions from a Question Bank. 5. User reviews the quiz information and questions. 6. User selects **Save**. 7. System validates the quiz data. 8. System creates the Quiz and associates it with the User. 9. System confirms that the Quiz has been created successfully.
Alternative Sequences / Flows	**A1 – Required information is missing:** System displays validation messages and does not create the Quiz until the required information is provided. **A2 – Invalid quiz information:** System displays an appropriate error message and allows the User to correct the data. **A3 – User cancels creation:** System cancels the operation and no Quiz is created.

2.3.2 Manage Quiz Questions

Item	Description
Use Case ID	UC13
Use Case Name	Manage Quiz Questions
Primary Actor	User
Secondary Actor	None

Item	Description
Description	As a User, I want to manage questions and answers in my Quiz so that I can build and maintain appropriate quiz content.
Preconditions	The User is authenticated and is the owner of the selected Quiz.
Postconditions	The Quiz contains the updated set of questions and answers in the specified order.
Normal Sequence / Flow	1. User opens one of their Quizzes for editing. 2. System displays the current Quiz questions. 3. User chooses to add, edit, remove, or reorder a Question. 4. When adding a Question, User can create a new Question or select an existing Question from a Question Bank. 5. For a newly created Question, User enters the question content, question type, and appropriate answers. 6. System validates the Question and Answer data. 7. User saves the changes. 8. System updates the Quiz and confirms the changes.
Alternative Sequences / Flows	**A1 – Invalid Question or Answer data:** System displays validation errors and does not save the invalid data. **A2 – User selects a Question from Question Bank:** System adds the selected Question to the Quiz without modifying the original Question. **A3 – User removes a reused Question:** System removes the relationship between the Quiz and Question but does not delete the original Question from the Question Bank. **A4 – User does not own the Quiz:** System denies the modification request.

2.3.3 Publish Quiz

Item	Description
Use Case ID	UC14
Use Case Name	Publish Quiz
Primary Actor	User
Secondary Actor	None
Description	As a User, I want to publish my Quiz so that other users can discover and participate in it.
Preconditions	The User is authenticated, owns the selected Quiz, and the Quiz has been created successfully.
Postconditions	The Quiz visibility is changed to Public and becomes available for discovery and participation by other users.

Item	Description
Normal Sequence / Flow	1. User opens one of their Quizzes. 2. User selects **Publish**. 3. System validates the Quiz and its questions. 4. System asks the User to confirm the publishing action. 5. User confirms. 6. System changes the Quiz visibility from Private to Public. 7. System makes the Quiz available in public browsing and search results. 8. System displays a successful publication message.
Alternative Sequences / Flows	**A1 – Quiz does not contain valid content:** System rejects publication and informs the User which information must be completed. **A2 – User cancels confirmation:** System keeps the Quiz private and makes no changes. **A3 – User does not own the Quiz:** System denies the publishing request. **A4 – Unpublish Quiz:** The owner may change a published Quiz back to Private. The system removes it from public discovery and prevents new public attempts.

2.4 Question Bank

2.4.1 Manage Question Bank

Item	Description
Use Case ID	UC15
Use Case Name	Manage Question Bank
Primary Actor	User
Secondary Actor	None
Description	As a User, I want to manage my Question Banks so that I can organize, store, and reuse questions when creating quizzes.
Preconditions	The User is authenticated.
Postconditions	The User's Question Bank data and contained Questions are created or updated successfully.
Normal Sequence / Flow	1. User opens the **Question Bank** function. 2. System displays the User's existing Question Banks. 3. User selects an existing Question Bank or creates a new one. 4. User enters or updates Question Bank information such as name, description, and visibility. 5. User adds, edits, or removes Questions within the selected Question Bank. 6. For each Question, User provides question content, question type, and related answers.

Item	Description
	7. System validates the entered data. 8. User saves the changes. 9. System updates the Question Bank and displays a success message.
Alternative Sequences / Flows	**A1 – Required information is missing:** System displays validation messages and asks the User to complete the required fields. **A2 – Invalid Question or Answer data:** System rejects the invalid data and asks the User to correct it. **A3 – Delete a Question Bank:** System checks whether the Question Bank or its Questions are currently being used by existing Quizzes before completing the deletion. **A4 – User attempts to modify another User's Question Bank:** System denies the request.

2.4.2 Reuse Question

Item	Description
Use Case ID	UC17
Use Case Name	Reuse Question
Primary Actor	User
Secondary Actor	None
Description	As a User, I want to reuse an existing Question from a Question Bank so that I can create quizzes more efficiently without recreating the same question.
Preconditions	The User is authenticated and is editing a Quiz. The selected Question is accessible to the User through their own Question Bank or a public Question Bank.
Postconditions	The selected Question is associated with the User's Quiz without changing the original Question data.
Normal Sequence / Flow	1. User selects **Add from Question Bank** while editing a Quiz. 2. System displays or allows the User to search available Questions. 3. User selects a Question. 4. System displays the Question details for review. 5. User confirms the selection. 6. System creates the relationship between the Quiz and the selected Question. 7. System places the Question in the Quiz according to the selected order. 8. System confirms that the Question has been added successfully.
Alternative Sequences / Flows	**A1 – No matching Question is found:** System informs the User that no Question matches the search criteria. **A2 – Question is not accessible:** System denies the request if the Question

Item	Description
	belongs to a private Question Bank owned by another User. **A3 – Question already exists in the Quiz:** System warns the User and prevents duplicate addition, if duplicate Questions are not allowed. **A4 – Original Question is later removed from the source Question Bank:** The system must preserve data consistency for Quizzes that already reference the Question according to the defined business rule.

2.5 System Administration

2.5.1 Moderate Content

Item	Description
Use Case ID	UC20
Use Case Name	Moderate Content
Primary Actor	Administrator
Secondary Actor	None
Description	As an Administrator, I want to review and moderate public quizzes and questions so that inappropriate or invalid content can be controlled within the system.
Preconditions	The Administrator is authenticated and has administrative privileges. The content to be reviewed exists in the system.
Postconditions	The selected content remains unchanged or an appropriate moderation action is applied. The system stores the updated content status.
Normal Sequence / Flow	1. Administrator accesses the content management function. 2. System displays quizzes and questions available for administration. 3. Administrator searches for or selects the content to review. 4. System displays the content details and related information. 5. Administrator reviews the content. 6. Administrator selects an appropriate moderation action. 7. System requests confirmation when necessary. 8. Administrator confirms the action. 9. System applies the moderation action and updates the content status. 10. System displays a success message.
Alternative Sequences / Flows	**A1 – No action is required:** Administrator finishes reviewing the content without making changes. **A2 – Remove or disable inappropriate content:** System removes the content from public access or changes its status so that other users can no longer access it. **A3 – Administrator cancels the action:** System makes no changes to the content.

Item	Description
	A4 – Content no longer exists: System informs the Administrator that the selected content is unavailable and returns to the management list.